Lab Exercise 9- Create Service in Kubernetes

Objective:

- Understand the syntax and structure of a Kubernetes Service definition file (YAML).

Prerequisites

- Kubernetes Cluster: Have a running Kubernetes cluster (locally using Minikube or kind, or a cloud-based service).
- kubectl: Install and configure kubectl to interact with your Kubernetes cluster.
- Basic Knowledge of YAML: Familiarity with YAML format will be helpful for understanding Kubernetes resource definitions.

Step-by-Step Guide

NodePort Service

To expose the Service on a port on each Node in the cluster, modify the Service type to NodePort.

Create a YAML file named ***service.yaml*** with the following content:

service.yaml

```
apiVersion: v1
kind: Service
metadata:
  name: nodeport-service
spec:
  selector:
    app: web
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
      nodePort: 30007 # A specific port in the range 30000-32767
  type: NodePort
```

Explanation:

- The primary difference from the ClusterIP Service is the addition of nodePort, which specifies the static port on each Node.
- type: Set to NodePort, exposing the Service on a specific port across all Nodes.

Apply this YAML to create the NodePort Service:

```
kubectl apply -f nodeport-service.yaml
```

Verify the Service:

```
kubectl get services
```

You should see the nodeport-service listed with a NodePort and details about the port exposed.

```
Microsoft Windows [Version 10.0.26200.8246]
(c) Microsoft Corporation. All rights reserved.

C:\Users\NK>(echo apiVersion: v1 & echo kind: Service & echo metadata: & echo   name: nodeport-service & echo spec: & echo   selector: & echo     app: web &
 echo   ports: & echo     - protocol: TCP & echo       port: 80 & echo       targetPort: 80 & echo       nodePort: 30007 & echo   type: NodePort) > nodeport
-service.yaml

C:\Users\NK>type nodeport-service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nodeport-service
spec:
  selector:
    app: web
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
      nodePort: 30007
  type: NodePort

C:\Users\NK>minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured


C:\Users\NK>kubectl apply -f nodeport-service.yaml
service/nodeport-service created

C:\Users\NK>kubectl get services
NAME              TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
kubernetes        ClusterIP   10.96.0.1       <none>        443/TCP        2m38s
nodeport-service  NodePort    10.100.204.78   <none>        80:30007/TCP   5s

C:\Users\NK>kubectl describe service nodeport-service
Name:                     nodeport-service
```

```
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured


C:\Users\NK>kubectl apply -f nodeport-service.yaml
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NAME              TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
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nodeport-service  NodePort    10.100.204.78   <none>        80:30007/TCP   5s

C:\Users\NK>kubectl describe service nodeport-service
Name:                     nodeport-service
Namespace:                default
Labels:                   <none>
Annotations:              <none>
Selector:                 app=web
Type:                     NodePort
IP Family Policy:         SingleStack
IP Families:              IPv4
IP:                       10.100.204.78
IPs:                      10.100.204.78
Port:                     <unset>  80/TCP
TargetPort:               80/TCP
NodePort:                 <unset>  30007/TCP
Endpoints:                10.244.0.2:80
Session Affinity:         None
External Traffic Policy:  Cluster
Internal Traffic Policy:  Cluster
Events:                   <none>

C:\Users\NK>minikube service nodeport-service --url
http://127.0.0.1:49765
```

```
C:\Users\NK>kubectl delete service nodeport-service
service "nodeport-service" deleted from default namespace

C:\Users\NK>kubectl get services
NAME         TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)   AGE
kubernetes   ClusterIP   10.96.0.1    <none>        443/TCP   5m10s
```